

Cell Specialisation

Quick reference sheet · Biology · keep handy for revision

The 3-step method

Every "explain how structure relates to function" question uses the same three steps — even for a cell you have never seen before:

1. Name the structural feature. 2. Say what it physically does (longer/more/thinner). 3. Link it to the cell's job.

1. Specialised animal cells

Cell	Key feature(s)	Why it helps
Sperm cell	Long tail; many mitochondria; acrosome (enzyme tip)	Swims to the egg; energy for swimming; digests through the egg membrane
Nerve cell	Long axon; branched endings; insulating layer	Carries signals long distances; connects to many cells; speeds up the signal
Muscle cell	Long fibre that contracts; many mitochondria	Shortening causes movement; energy for contraction

2. Specialised plant cells

Cell	Key feature(s)	Why it helps
Root hair cell	Long hair-like extension	Large surface area for absorbing water & mineral ions from soil
Xylem cell	Hollow tube, no contents; lignin-thickened walls	Water flows through easily; strong enough to support the plant
Phloem cell	Sieve tube with pores; companion cell alongside	Sugars flow through; companion cell provides energy for active transport of sugars

3. Worked example (unseen cell)

"A cell lining the small intestine has many finger-like projections on one surface. Suggest how this helps it absorb nutrients."

1. Feature: finger-like projections. 2. What it does: greatly increases surface area. 3. Link: more surface area means more nutrients absorbed at once.